

## 2. Merged PBP and Participation Data Pipelines

### Overview

To give the AI context about what defensive shells actually work, we had to stitch together two completely separate NFL datasets. The standard Play-by-Play (PBP) dataset contains the math (EPA), but the Next Gen Stats (NGS) Participation dataset contains the context (defensive coverages, box counts, offensive personnel).

### Code Changes

- **Data Engineering (Pandas):** Engineered a Python script utilizing the `nfl_data_py` library. We isolated the 2024 PBP data and executed an inner join with the NGS Participation CSV on the `old_game_id` and `play_id` columns.
- **Dataset Generation:** This resulted in a unified `master_pbp_participation_2024.csv` file, serving as the ultimate ground-truth dataset where every specific coverage type is permanently linked to the exact EPA it generated.

### Future Roadmap

Currently, this is a static script run on a local CSV. The next step is to automate this pipeline using GitHub Actions to fetch, clean, merge, and store the latest NFL data automatically every Tuesday morning during the regular season, ensuring the AI's models learn dynamically week-over-week.